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Determining the effect of olfactory receptors on the survival to *Pseudomonas entomophila* infection in *Drosophila melanogaster*

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Resumo

O estudo do ciclo das relações entre patógeno e hospedeiro sempre foi importante para embasar o conhecimento dos mecanismos imunes por de trás do combate a doenças. A resposta imune inata, presente em ambos vertebrados e invertebrados, é a mais primordial e não depende de uma memória imune, ao contrário da resposta imune adaptativa, que apenas consta em vertebrados. *Drosophila melanogaster*, ao ter apenas o primeiro tipo de resposta imune citado, permite-nos isolar o sistema imune inato, facilitando o estudo do mesmo.

Em *Drosophila* temos quatro tipos principais de resposta imune: O evitamento comportamental, onde o indivíduo fisicamente reage de várias formas para evitar a infeção; As barreiras epiteliais, que ajudam a travar a entrada do patógeno para dentro do organismo; A imunidade humoral que vai produzir péptidos anti microbiais e outros produtos no intestino, que vão funcionar de forma semelhante a antibióticos; Finalmente, existe a imunidade celular que produz hemócitos, células imunes, com funções especializadas de forma a travar o alastramento da infeção. Esta produção de hemócitos é cessada no estágio larval, ficando o adulto com as células remanescentes após a pupação.

Além do sistema imune, outro mecanismo importantíssimo na mediação de ameaças potenciais de patógenos é o sistema olfativo. Normalmente o comportamento deste sistema é associado com alterações comportamentais que levam ao evitamento da infeção, no entanto, mais recentemente, estudos têm provado que na verdade este sistema pode também interagir de forma direta com os canais imunes. Concretamente, este sistema consegue sinalizar para que células imunes comecem a ser produzidas ao detetar certos estímulos olfativos, conferindo assim um *boost* imunitário ao hospedeiro. Além disso, consegue também iniciar uma cascata que acaba por levar à ativação de uma protéase responsável pelo declínio do sistema olfativo e também pela regulação da saúde neural numa idade avançada. No entanto, o conhecimento sobre o real papel deste sistema na imunidade é ainda quase que completamente desconhecido.

No adulto de *Drosophila* existem dois apêndices principais onde ocorre a olfação: a antena e os palpos maxilares. Estas estruturas são cobertas por vários elementos porosos denominados de sensilas e é nelas onde se encontram os neurónios recetores olfativos. Estes neurónios estão cobertos por recetores olfativos que detetam moléculas chamadas de odorantes. Para serem funcionais, os recetores olfativos estão acoplados a um co-recetor estável chamado de ORCO.

Nesta tese investigámos o possível efeito destes recetores olfativos durante a resposta imune. Para tal, recorrendo à tecnologia de RNAi, foram criadas linhas sem a expressão do gene do recetor em 43 recetores olfativos diferentes. Essas linhas, além de não expressarem o recetor olfativo, tinham ainda um sistema UAS/Gal4/Gal80 associado para que fosse possível controlar quando essa expressão ocorria ou não, evitando afetar o desenvolvimento dos indivíduos até ao estágio adulto, quando as experiências ocorreram.

Para inferir a resposta imune das diferentes linhas, infetámos moscas de ambos os sexos com *Pseudomonas entomophila*, uma bactéria patogénica gram-negativa, e medimos a sobrevivência. Verificámos que ocorreu uma variação significativa nas curvas de sobrevivência, com uma grande amplitude tanto em machos quanto em fêmeas. Esses resultados mostram que ou a falta de expressão dos genes influencia o comportamento do sistema imune nestes indivíduos, ou então, essa falta de expressão faz com que o *fitness* das moscas seja diferente entre si.

Com base nos resultados obtidos, especulamos que a causa desta variação poderá ser uma maior ou menor expressão de péptidos antimicrobianos. Esta hipótese tem sido descoberta como estando por trás de outros mecanismos imunes, inclusive num estudo que foca em recetores olfativos e neurónios recetores olfativos. Outra hipótese seria o mecanismo de RNAi em si, já que foi provado que a temperatura afeta a expressão dos recetores olfativos, fazendo com que o mecanismo seja mais efetivo em genes sub-expressos do que em genes sobre-expressos.

Uma vez que tínhamos as curvas de sobrevivência em ambos machos e fêmeas, tentámos compreender se existia algum dimorfismo sexual na resposta imune. De facto, esse dimorfismo pôde ser encontrado em algumas das linhas estudadas. A principal interpretação biológica por de trás deste resultado seria uma imunidade específica do sexo, que é plenamente aceite pela sociedade científica em várias outras situações. Outras possíveis interpretações seriam um impacto específico do sexo no sistema RNAi, o que já foi provado que ocorre em alguns genes relacionados com a determinação sexual. Outra hipótese seria algo ao nível do sexo conferir uma expressão diferenciada aos genes, como no caso do recetor olfativo responsável por detetar hormonas de cortejamento, ou até mesmo o comportamento social diferencial entre machos e fêmeas.

Ainda com os dados de sobrevivência, através de uma análise de *clustering*, foi possível isolar as linhas com o comportamento mais divergente em machos e fêmeas. Curiosamente, o co-recetor ORCO foi selecionado como sendo uma linha divergente em machos. Também relevante, as linhas selecionadas em machos e fêmeas, são, na sua grande maioria, diferentes.

Uma vez que conseguimos isolar as linhas mais divergentes decidimos medir o *output* reprodutivo em fêmeas. Esta experiência ocorreu de forma semelhante à primeira, onde infetámos moscas com *P. entomophila*, no entanto, após a infeção além de medirmos a mortalidade, medimos também o número de pupas em cada tubo. Observámos que havia uma grande variação no *output* em todas as linhas. Além disso, verificámos que na maior parte das linhas a maior pupação ocorreu nos dias 7 e 8, podendo indicar algum tipo de mecanismo de sobrevivência, como por exemplo, a diminuição da oviposição durante os primeiros estágios da infeção.

Com isto, ao tentar inferir possíveis *trade-offs* que tenham ocorrido durante a infeção nas linhas divergentes, elaborámos algumas hipóteses que poderiam ter ocorrido. Primeiramente, a hipótese de ser devido a conservar energia para combater a infeção, como explicado em cima, a oviposição seria reduzida durante o início da infeção para conservar energia, aumentando uma vez que a infeção estivesse combatida. Em segundo lugar seria a hipótese de que linhas com menor sobrevivência teriam mais dificuldade em detetar o patógeno do que as restantes linhas, fazendo com que consumissem mais bactéria, levando a uma maior concentração da mesma no corpo, dificultando o combate à infeção. Finalmente, poderá ser que as moscas com menor mortalidade tenham simplesmente um melhor sistema imune, combatendo a infeção mais rapidamente, conseguindo pôr mais ovos.

Para, então, tentarmos inferir se existe alguma correlação entre a fecundidade e a mortalidade, traçámos uma correlação de *Pearson* entre ambas. Surpreendentemente, os resultados mostram que não existem qualquer relação entre elas, impossibilitando uma resposta mais preliminar sobre qual, ou, até, se alguma das hipóteses se aproxima mais da realidade.

Resumindo, nesta tese de mestrado tentamos entender qual os efeitos do olfato na resposta imune em *Drosophila melanogaster*. Concluímos que diferentes linhas de recetores olfativos sem expressão sobrevivem de forma diferente à infeção de *Pseudomonas entomophila* e que existe dimorfismo sexual

em algumas destas linhas. Pudemos ainda isolar as linhas com uma resposta mais divergente e medir o *output* reprodutivo destas linhas, em fêmeas. Finalmente, cruzámos os resultados de sobrevivência com os resultados de *output* reprodutivo e verificámos que não existe qualquer correlação entre os mesmos. Os resultados aqui obtidos mostram que existe ainda muito a se estudar no que toca à função do olfato na imunidade. Esta tese procurou elucidar aquilo que é conhecido em relação à função dos recetores olfativos no sistema imune, no entanto, ainda existem muitas lacunas que espero serem preenchidas ao longo dos próximos anos.

Palavras-chave: *Drosophila melanogaster*; Imunidade; *Pseudomonas entomophila*; *Output* reprodutivo; Olfato

Abstract

Olfaction plays a critical role in the mediation of interactions between hosts and pathogens. Such role is primarily linked to the evasion of microorganisms through several behavioural adaptations. Yet, more recently, studies have found that olfactory signalling may directly influence immune responses in *Drosophila melanogaster*. Although such findings have proven to be enlightening when it comes to what we know about the effect of olfactory receptors in immunity, the knowledge on the extent of that effect is still limited.

In this MSc we try to expand that knowledge by analysing the consequences of knocking down olfactory receptors in *D. melanogaster* and infecting them with *Pseudomonas entomophila*, a pathogenic gam-negative bacterium. We employed RNAi to target 43 olfactory receptor lines and performed multiple survival assays using both males and females, noting differences in overall survival rates in both sexes. We also found that there was sexual dimorphism upon surviving infection in some of the lines. With the survival rates we obtained, and resorting to clustering analysis, we were able to select the most diverging lines for both males and females.

To further examine the boarder ramifications of olfactory receptor knockdown, we assessed the reproductive output of the selected divergent lines in females. We aimed to uncover any trade-offs or reproductive strategy in those lines and found that there is a big variation among all lines. Finally, we found that no consistent correlation could be identified between mortality and reproductive output across the selected lines.

Our results highlight the diversity of behaviours that can be found in infecting knockdown lines of olfactory receptors, throughout this thesis it is also shown how information on olfactory receptors is lacking. The true extent of the roles that olfaction has are still widely unknown and misunderstood, with this study we aimed to shed some light on the matter.

Keywords: *Drosophila melanogaster*; Immunity; *Pseudomonas entomophila*; Reproductive output; Olfaction

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List of Abbreviations

AMP	Antimicrobial peptide
AL	Antennal Lobe
BDSC	Bloomington *Drosophila* Stock Center
CA	Calyx
cmix	*intersex* gene
D. melanogaster	*Drosophila melanogaster*
dsRNA	Double-stranded RNA
dsx	*doublesex* gene
GABA	Gamma-aminobutyric acid
GPCR	G-protein-coupled receptor
GR	Gustatory receptor
HIFα	Hypoxia-inducible factor alpha
IR	Ionotropic receptor
LB	Life broth
LH	Lateral horn
OR	Odorant/olfactory receptor
ORCO	Olfactory receptor co-receptor
ORN/OSN	Olfactory receptor neuron/Olfactory sensory neuron
P. entomophila	*Pseudomonas entomophila*
PN	Projection neuron
RNA	Ribonucleic acid
ROS	Reactive oxygen species
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
UAS	Upstream activating sequence
VDRC	Vienna *Drosophila* Resource Center

1. Introduction

Host-pathogen interactions are parasitic symbiotic relationships where one strives to overcome the other (Tan & Ausubel, 2002). Such interactions often focus on the infection of a host by different types of pathogens. This will lead to one being positively benefited, usually the pathogen, and the other suffering the consequences, often, the host. The back and forth of infecting and being infected creates a co-evolutionary cycle where the host develops mechanisms to avoid and surpass infection, such as the creation of an immune system (Burdon et al., 2013). Similarly, pathogens evolve to better avoid the barriers put up by the host, facilitating the spread of the disease to new individuals (Bliven & Maurelli, 2016; Burdon et al., 2013). One example of a successful pathogenic evolution was the last severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic, where the increase in viral load has been appointed as a cause for a growth in transmission between humans (Puhach et al., 2022). Boosting population immunity against SARS-CoV-2 was only possible due to the accumulated knowledge on immunity (Puhach et al., 2022). Thus, understanding immune mechanisms not only sheds light on how hosts counteract pathogens but can also have far-reaching implications for protecting human health.

There are two types of immune response: adaptive immune response and innate immune response (Marshall et al., 2018). Adaptive immune response occurs exclusively in vertebrates and its mechanism is based on the activation of B and T lymphocyte cells (Arneth, 2021; Bonilla & Oettgen et al., 2010). Innate immune response, the one we focus on in this thesis, is the most primordial type of immune response, common to both vertebrates and invertebrates. Innate immunity is mainly a non-specific way of dealing with pathogens (Arneth, 2021; Netea et al., 2016). Unlike adaptive immune response, this type of immune response does not rely on immunologic memory, thus it will not have an immune boost and accelerated response when encountering the same pathogen a second time (Marshall et al., 2018; Figure 1.1). Even so, some studies have shown that innate immune responses can be, to some extent, specific in the presence of pattern recognition receptors. These receptors are capable of recognizing conserved microbial structures and also to activate essential pathways in immune response, which confers some specificity to the mechanism (Arneth, 2021; Lu et al., 2019).

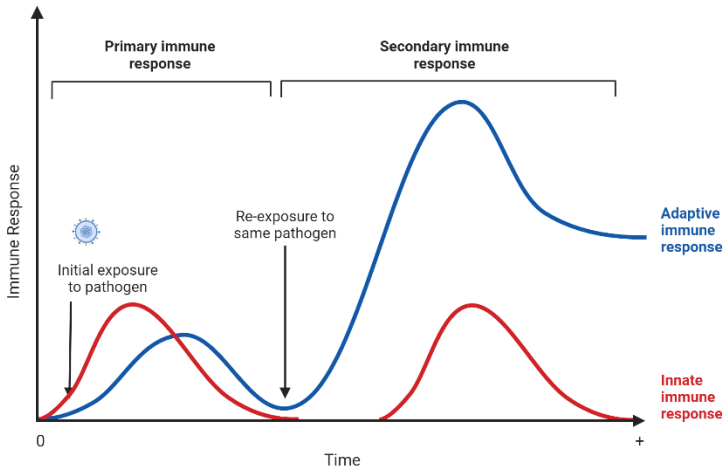


Figure 1.1 – Adaptive and innate immune responses over time and after re-exposure. Adaptive immune response increases when there is re-exposure, showing specificity and immune memory. Innate immune response remains the same due to being non-specific. Created with BioRender.com.

Fuelled by its extensive genetic toolkit, *Drosophila melanogaster* has long been a valuable model for studying immune responses and gene expression (Moraes & Montagne, 2021). Fruit flies have a short life cycle and are easy to maintain, making them ideal for study. The complexity of mammalian models makes them particularly attractive for those who prefer a utilitarian approach to science, as they more closely mimic human physiology. The advantage of *Drosophila* as a model is that it is significantly simpler than any vertebrate as it relies solely on innate immune mechanisms that are highly conserved across species (Yu et al., 2022). While humans have 46 chromosomes, *Drosophila* has only eight (Kaufman, 2017; Trask, 2002). Despite these differences, approximately 75% of genes that are responsible for causing diseases in humans are thought to have functional homologs in *Drosophila*, allowing for meaningful comparisons between the two species in immune studies (Ugur et al., 2016; Moraes & Montagne, 2021).

The olfactory system plays a critical role in mediating host-pathogen interactions, it enables detection of environmental cues such as potential threats. This sensory system is crucial for survival, as it helps organisms adapt to their surroundings and respond appropriately to environmental challenges, including those posed by pathogens (Dweck et al., 2015; Kohlmeier et al., 2021). In order to fully dwell into the role of olfaction in such interactions we must comprehend how the immune system in *Drosophila* works.

Drosophila immune response can be divided into four main types of mechanisms. First, behavioral avoidance helps prevent infections through behaviours, such as changes in oviposition strategies in the presence of parasitoid wasps (Lynch et al., 2016) or a decrease in motivation to forage for food after prior exposure to *Drosophila* C Virus (Vale & Jardine, 2016). Second, epithelial barriers—comprising the epidermis, gut, and trachea—act as physical defences against pathogens, supported by a protective chitinous matrix. The production of antimicrobial peptides (AMPs) and reactive oxygen species (ROS) in the gut help surviving infections (Lemaitre & Hoffman, 2007). Third, systemic humoral immunity involves the fat body, an organ that functions analogously to adipose tissue, liver and immune system in vertebrates (Figure 1.2B). It serves as the primary fat storage site and secretes immune peptides, including AMPs, into the hemolymph. Genes encoding AMPs and other immune-related products are primarily regulated by the Toll and immune deficiency (Imd) pathways (Lemaitre & Hoffmann, 2007; Lu et al., 2019; Yu et al., 2022, Musselman et al., 2013). Lastly, cellular immunity relies on hemocytes, which employ strategies such as phagocytosis and melanisation to neutralise pathogens (Banerjee et al., 2019).

Cellular immunity in *Drosophila* has three primary immune cell lineages, also known as hemocytes. The most abundant are plasmatocytes, constituting 90-95% of all blood cells. The primary function of plasmatocytes is to phagocytose microorganisms and apoptotic cells. Crystal cells play a pivotal role in the melanisation pathway, where melanin is crucial for preventing hemolymph loss (Cooper et al., 2019; Duvic et al., 2002; Makhijani & Brükner, 2012; Meister & Lagueux, 2003). Lamellocytes are specialised immune cells that differentiate exclusively in response to parasitoid wasp infection (Makhijani & Brükner, 2012; Meister & Lagueux, 2003). Lamellocytes are formed through differentiation of plasmatocytes in the lymph gland (Grigorian & Hartenstein, 2013).

Hematopoiesis is the process by which stem cells differentiate into hemocytes, driven by lineage-determining transcription factors and the activation of regulated gene expression pathways (Jaitin et al., 2015). In *Drosophila*, hematopoiesis occurs in two distinct waves at different life stages (Morin-Poulard et al., 2021). The first wave takes place during embryogenesis, occurs in the embryonic mesoderm, producing hematopoietic progenitors known as prohemocytes, which later differentiate into functional immune cells (Makhijani & Brückner, 2012). The second wave, during the larval stage takes place in

two distinct hematopoietic niches: the lymph gland and hematopoietic clusters (Morin-Poulard et al., 2021).

There are two main hematopoietic locations in *Drosophila*. The lymph gland serves as the primary hematopoietic organ in *Drosophila* larvae and has been extensively studied in immune and functional research (Lan et al., 2020; Lemaitre & Hoffmann, 2007; Yu et al., 2022). It consists of multiple lobes, and by the third-instar larval stage, the primary lobe can be divided into distinct zones. The medullary zone, located centrally, primarily contains pro-hemocytes, while the cortical zone houses mature blood cells. Another critical region is the posterior signalling centre, which regulates homeostasis (Lan et al., 2020; Luo et al., 2020). In response to immune challenges, such as parasitoid wasp infections, immune cells in the lymph gland may undergo transdifferentiation. During metamorphosis, the lymph gland degenerates, releasing stored plasmatocytes and crystal cells into circulation (Lan et al., 2020).

While the lymph gland has historically been considered the primary site of hematopoiesis (Lemaitre & Hoffman 2007, Lan et al., 2020), some studies have expanded this knowledge to include additional niches, such as dorsal sessile clusters (Márkus et al., 2009; Leitão & Sucena, 2015; Figure 1.2A). Sessile clusters are zones where hemocytes attach to the epidermis of the cuticle, forming clusters in each segment of the larva. It is known that the modus operandi of sessile clusters is slightly different from the lymph gland one. A key difference lies in the mechanisms of crystal cell differentiation. In the lymph gland, differentiation is regulated by distinct pathways, whereas in sessile clusters, it is controlled by the Lozenge gene, which relies on Notch signalling and contact-dependent interactions with its ligand, Serrate (Duvic et al., 2002; Leitão & Sucena, 2015).

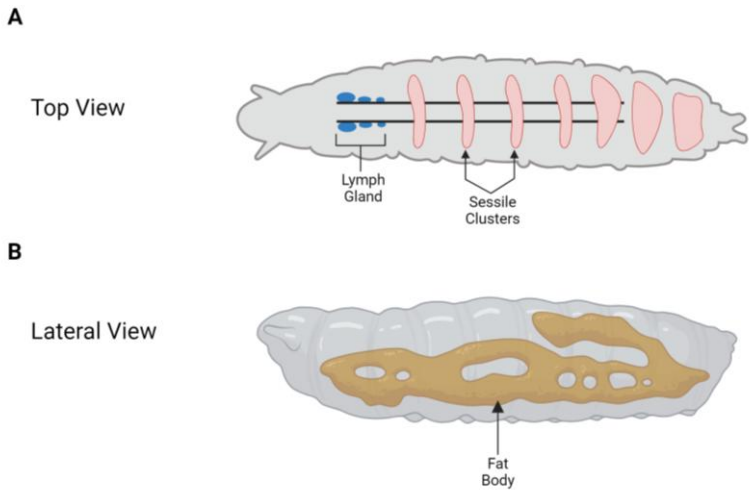


Figure 1.2 – Larval hematopoietic organs and fat body in *Drosophila* larva. (A) Scheme from a top perspective that shows the lymph gland and sessile clusters and **(B)** Lateral view that shows the fat body. Created with BioRender.com.

Despite the extensive research on *Drosophila* immune response, studies focusing on the adult innate immune system remain scarce. The transition from larval to adult immune function is marked by the persistence of hemocytes generated during larval hematopoiesis. These cells maintain their immune functions (Lan et al., 2020). Boulet et al. (2021) further demonstrated that while most of the adult blood cells are mature, a subset of pluripotent stem cell-derived hemocytes retains the ability to perform hematopoiesis to some extent. Interestingly, the same study revealed an unexpected similarity between adult and embryonic blood cells, distinguishing them from larval hemocytes (Boulet et al., 2021).

Although the prevailing view has been that adult blood cells are derived entirely from embryonic and larval hematopoietic processes, Ghosh et al. (2015) challenged this notion, suggesting that adult plasmatocytes proliferate in response to *Escherichia coli* infection and that blood cell progenitors capable of differentiation persist in adulthood. However, Bosch et al. (2019) found no corroboration for those claims having not found any sign of adult blood cell differentiation or proliferation in their work. This debate underscores the complexity of *Drosophila* hematopoiesis and the gaps in our understanding of its regulation in adulthood (Boulet et al., 2021).

Similarly, another feature conserved between the larval and adult stages is the fat body, which continues to play critical roles in immunity, metabolism, and endocrine signalling, ensuring the organism's ability to adapt to environmental challenges and sustain homeostasis throughout its life cycle (Vanha-aho et al., 2015; Figure 1.3). The activation of the production of AMPs in the adult fat body can often be caused by the sensing of environmental cues, highlighting the importance of the sensory system in the mediation of the immune response (Ebrahim et al., 2015; Shoofs et al., 2014; Hückesfeld et al., 2016).

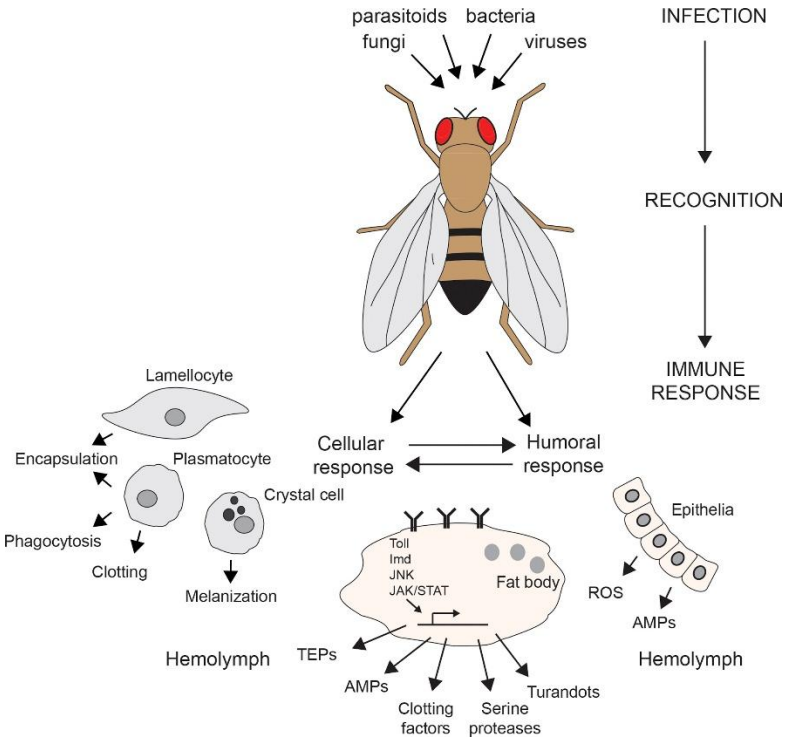


Figure 1.3 – Types of immune response in the *Drosophila* adult. Types of immune response that persist in the adult after the larval phase in *Drosophila*. Adapted from Vanha-aho et al. (2015).

In adult *D. melanogaster*, there are two primary sensory appendages responsible for olfaction: the antennae and the maxillary palps. Both structures are covered with hair-like sensilla, specialised units involved in detecting odours (Barish & Volkan, 2015; Wilson, 2013). Each sensillum hosts between one and four olfactory receptor neurons (ORNs) and can be classified into four types—trichoid, intermediate, coeloconic, or basiconic—based on its structure and the arrangement of cuticular pores (Gonzales et al., 2021; Gomez-Diaz et al., 2018). The cell bodies of ORNs, along with a cluster of supporting cells, are situated at the base of the sensilla. The dendrites of the ORNs extend through the sensillar protrusion at the top, where direct contact with the environment occurs, allowing odorant detection (Gonzales et al., 2021). The structural classification of a sensillum influences both the dendritic branching patterns and the odour selectivity of its ORNs. This suggests that sensilla may play

a direct role in shaping olfactory function, as proposed by Gonzales et al. (2021), although this hypothesis remains to be experimentally validated.

When describing *D. melanogaster* brain, the main olfactory processing centre can be found in the antennal lobe. This region is organised into spherical structures called glomeruli, where axon terminals from ORNs form synapses with projection neurons (PNs) (Gruber et al., 2023; Figure 1.4). Each ORN is typically activated by multiple odour ligands, with its activation determined by factors such as odour identity, concentration, and the rate of change in concentration (Wilson, 2013). Despite the enormous existing range of ORNs, this type of neuron tends to express a single type of receptor, with neurons of the same type converging in a single glomerulus (Wilson, 2013; Vosshall et al., 2000). The projection neurons then relay this processed information to higher brain centres for further integration and response (Barish & Volkan, 2015; Wilson, 2013).

The ability of ORNs to transmit signals to the antennal lobe and higher brain centres relies on the receptors expressed in these neurons. These receptors act as the molecular interface between environmental odorants and the neural circuits that process olfactory and taste information (Gomez-Diaz et al., 2018). In *D. melanogaster*, three main receptor families mediate this sensory detection: Gustatory Receptors (GRs), Ionotropic Receptors (IRs), and Odorant Receptors (ORs).

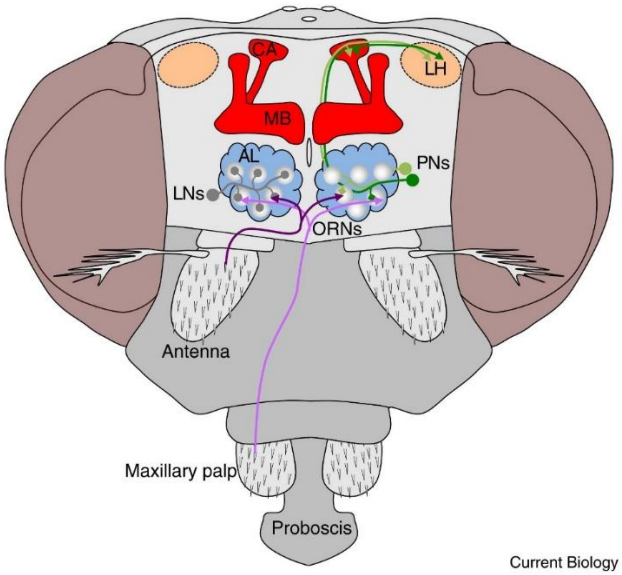


Figure 1.4 – Olfactory pathway in *Drosophila*. Olfactory receptor neurons (ORNs) at the maxillary palps project their axons to the glomeruli in the antennal lobe (AL). Once the information has been processed by local interneurons (LNs) the input is taken to the calyx (CA) of the mushroom body (MB) and the lateral horn (LH) by projection neurons (PNs). Adapted from Perisse et al. (2013).

Firstly, GRs, primarily associated with the insect's palate, are essential for contact chemosensation. GRs are responsible for detecting non-volatile compounds like sugars and bitter substances, playing a critical role in taste perception and feeding behaviour (Montell, 2009). Second, IRs, which are thought to be involved in various sensory modalities, including chemosensation and thermosensation, with some studies suggesting their potential association with hearing (Ni, 2021). These receptors contribute to detecting water-soluble odorants and other non-volatile compounds, complementing the role of ORs in olfactory perception (Gomez-Diaz et al., 2018; Ni, 2021). Finally, Odorant Receptors ORs, also referred to as olfactory receptors, are transmembrane proteins that specialise in detecting volatile odorants. ORs form heteromeric complexes comprising an odorant-specific subunit and a conserved co-receptor known

as ORCO (Saberi & Seyed-Allaei, 2016; Figure 1.5). These receptors are unique in their dual functionality: while primarily classified as metabotropic receptors belonging to the G-protein-coupled receptor (GPCR) family, they also exhibit ionotropic properties, displaying ionic permeability to Ca^{2+}, Na^{+}, and K^{+}. Structurally, ORs feature an intracellular N-terminus and an extracellular C-terminus (Gomez-Diaz et al., 2018).

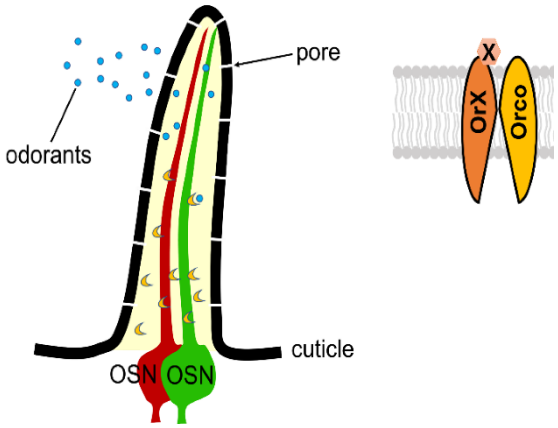


Figure 1.5 – Sensilla and olfactory receptor structure. Olfactory sensory neurons (OSNs) are located in porous sensilla in the maxillary palps and antenna. OSNs have receptors in their dendrites, those receptors are called Olfactory receptors (ORs). In order to be able to sense odorants, ORs bind to a stable co-receptor called ORCO. Adapted from Ha & Smith (2022).

The *D. melanogaster* genome contains 60 genes encoding olfactory receptors, resulting in 62 OR proteins, with two of these—Or46a and Or69a— being produced through alternative splicing (Gomez-Diaz et al., 2018). This diversity in ORs enables flies to detect and respond to a wide array of volatile compounds, highlighting their central role in olfactory-driven behaviours and environmental adaptation.

Traditionally, sensory receptors, including ORs, have been associated with behavioural avoidance mechanisms in response to environmental threats, such as pathogens. For instance, Shoofs et al. (2014) and Hückesfeld et al. (2016) investigated the role of hugin-expressing neurons, which function as the bitter taste processing centre in the brain. Their findings revealed that these neurons not only suppress other sensory modalities but also induce behavioural changes, such as reduced food intake and increased locomotion.

Additionally, research has shown that both adult and larval *Drosophila* possess dedicated olfactory circuits for detecting odours emitted by *Leptopilina* parasitoid wasps. Larvae only need a single receptor, Or49a, to identify iridomymercin. On the other hand, adults need both Or49a and Or85f to detect actinidine and nepetalactol. Interestingly, while olfactory activation occurred in both life stages, behavioural avoidance was observed only when visual cues were also present (Ebrahim et al., 2015).

The studies mentioned underscore the critical role of sensory circuits, including olfactory pathways, in detecting environmental and pathological threats and, in some instances, driving behavioural avoidance to enhance survival (Shoofs et al., 2014; Hückesfeld et al., 2016; Ebrahim et al., 2015). However, recent studies have begun to challenge this behavioural-centric view by uncovering direct connections between sensory receptors and molecular immune pathways. These emerging findings suggest that olfactory perception may play a more integrated role in coordinating immune responses, broadening our understanding of sensory-immune interactions.

In their 2020 study, Madhwal et al. explored the interplay between olfactory perception and immune function in *D. melanogaster*, focusing on the differentiation of lamellocytes—specialised blood cells that respond to parasitic wasp infections. They found that olfactory stimulation of larvae by parasitoid wasp odours, detected through the olfactory receptors Or42a and Or49a, triggers the neuronal release of gamma-aminobutyric acid (GABA). This GABA is subsequently utilised by hematopoietic progenitor cells, where its catabolism stabilises the hypoxia-inducible factor alpha (HIFα) protein, known as Sima in *Drosophila*. The stabilisation of Sima is essential for initiating lamellocyte differentiation, providing larvae with an immune boost and improving their chances of survival against parasitic threats (Madhwal et al., 2020).

Another study where olfactory receptors have a main role in immune response, Chihara et al. (2014) investigated the role of caspase-3-like activity in the ageing *Drosophila* brain, focusing on olfactory receptor neurons (ORNs). They observed age-dependent activation of caspase-3-like proteases, particularly in Or42b and Or92a neurons, which are essential for innate attraction to food-related odours. This activation was associated with a decline in olfactory function, suggesting that caspase activity contributes to sensory deterioration during ageing.

Building upon this, Takeuchi et al. in 2022 explored the non-apoptotic functions of caspase-3-like proteases in *Drosophila*. They demonstrated that sublethal activation of these caspases plays a role in neuronal remodelling and synaptic plasticity, processes crucial for maintaining neural circuit function. Their findings indicate that caspase-3-like activity, beyond its role in apoptosis, is integral to the dynamic regulation of neuronal structures, thereby influencing behaviour and neural health.

More recently, Mazariegos et al. (2024) presented findings in a preprint demonstrating that *Drosophila* larvae rely on their taste system to sense bacterial peptidoglycans in their environment. This detection plays a crucial role in modulating the activity of their cellular immune system, highlighting a direct link between sensory perception and immune responses in larvae. Although this study focuses on taste rather than olfactory circuits, it underscores the broader role of sensory systems in immune modulation.

The aforementioned studies demonstrate the significant role of sensory circuits, including olfactory pathways, in detecting environmental and pathological threats and modulating immune responses. However, the specific contribution of olfactory receptors (ORs) to immune function and survival in *D. melanogaster* remains largely unexplored.

In this study, we investigate the role of specific olfactory receptors in *D. melanogaster* by infecting OR knockdown lines with *Pseudomonas entomophila* to assess their impact on survival during infection. We expected for some of the lines to have a differential reaction to the infection, as shown in aforementioned in previous studies. Based on the data obtained, we performed a cluster analysis and identified lines that had extreme responses to the infection. We also found that, in some lines, there is a sexual dimorphic response to the infection. To our knowledge, this type of analysis has never been done before and should build onto knowledge of the role of ORs on immune infection. To explore potential trade-offs associated with differential mortality, we measured the fecundity of these lines under infection conditions. We observed that there was a weak correlation between fecundity and mortality, something that can lead us to find more on how infection of knockdown OR lines can influence the oviposition behaviour in females.

2. Materials and Methods

2.1 Genetic System for Olfactory Receptor Downregulation

To investigate the role of olfactory receptors (ORs) in survival following bacterial infection, we employed the UAS/Gal4/Gal80 system to downregulate specific OR and ORCO genes via RNA interference (RNAi). This system uses Gal4, expressed under the control of a tissue- or protein-specific promoter (driver), in combination with an upstream activating sequence (UAS). When Gal4 binds to the UAS, it induces the expression of the target gene—in this case, OR or ORCO—as a double-stranded RNA (dsRNA), which is subsequently degraded through RNAi mechanisms (Weaver et al., 2020; Brand & Perrimon, 1993).

To regulate this system temporally, we used Gal80, a temperature-sensitive repressor of Gal4 activity. At 18°C, Gal80 effectively inhibits Gal4, preventing activation of the UAS/Gal4 system during fly development. At higher temperatures, such as 29°C, Gal80 repression is reduced, allowing Gal4-mediated transcription and RNAi-mediated knockdown of the target gene to occur (Weaver et al., 2020; Eliason et al., 2018).

2.2 Crossing Scheme and Maintenance Conditions

In order to obtain the OR knockdown genotype fly lines expressing dsRNA for RNAi targeting OR-specific genes under UAS control were crossed with flies carrying a tubulin promoter and a Gal80/Gal4 system (*tub-GAL80ts; tub-GAL4/TM6*).

The parental generation was maintained at 25ºC, where crosses and oviposition occurred. Following oviposition, the egg-lay content was transferred to 18ºC throughout development. Both maintenance and experimental conditions consisted of 60% humidity, a 12:12 hour light-dark photoperiod, and a molasses, sugar, cornflour, beer yeast, agar, water and 10% nipagin nutrient medium, ordered from Champalimaud (www.fchampalimaud.org).

2.3 Stocks

Lines used for this study were purchased from the Bloomington *Drosophila* Stock Center (BDSC; bdsc.indiana.edu) and have all been previously described in Perkins et al., 2015. Initially, lines from both the Vienna *Drosophila* Resource Center (VDRC) and BDSC were screened, targeting all 60 olfactory receptor (OR) genes, splitting them across blocks. However, due to variability in results and challenges in interpreting the data, the results presented focused exclusively on a subset of 43 lines from BDSC that target olfactory receptors on chromosome 2 and share the same genetic background.

2.4 Bacteria preparation and infection protocol

We used *Pseudomonas entomophila* to infect the knockdown lines. *P. entomophila* is a gram-negative bacterium and a natural pathogen of *D. melanogaster* that can infect the host through ingestion. The pore-forming toxin produced by the bacteria inhibits the translation in the intestine, blocking epithelium renewal (Nonaka et al., 2020; Dieppois et al., 2014). In *Drosophila* adults the Pore-forming toxin effectors are fought with a cross-linked Droso-crystallin protein, which is impermeable to macromolecules, such as the pore-forming toxin (Nonaka et al., 2020).

The bacteria after having been harvested from an isolated colony was submerged in a glycerol medium and maintained at a -80ºC freezer. During each infection round, the stored bacterium was plated in LB and rifampicin plates. Rifampicin is an antibiotic that the strain of *P. entomophila* in use is resistant to, therefore, reducing the risk of other microorganisms contaminating the sample. The plate was kept at 29ºC for about one day and a half. To prepare a starter for the bacteria growth, we added an isolated to an autoclaved vial that contained 5mL of liquid LB, and incubated it in an orbital shaker, for a period of 8 hours. Afterwards, the initial solution was poured into an 2000mL erlenmeyer and 500mL-1000mL of liquid LB were added for an overnight incubation in the same orbital shaker.

The following day the bacteria solution was centrifuged at 4ºC, 4600 rpm during 15 minutes in a Himac CR22N high-speed refrigerated centrifuge. Next, the supernatant was decanted, and the pellet resuspended to adjust the concentration to OD600=100. Finally, the suspension was diluted with a 2% sucrose solution in a 1:1 ratio and added food colouring at a final concentration of 2%.

The flies for infection were subjected to a fasting period of 24 hours prior to the infection to ensure adequate consumption of the bacterial suspension. *D. melanogaster* were infected in 90mm diameter bottles that contained a circular filter paper (whatman) that had absorbed the bacterial solution and a wet cotton pad in the lid to provide a hydration source throughout the 3-hour infection at 29ºC.

2.5 Phenotyping

2.5.1 Mortality Protocol

To measure mortality in olfactory receptor (OR) knockdown flies, between 30 and 70 individuals of each sex were separated and placed in infection bottles at 29°C for 3 hours, we also checked for blue colouring in their abdomen, ensuring all flies had the opportunity to consume the bacterial suspension. After the infection period, in order to follow the mortality, create replication and ensure fly density is the same, 10 flies of the same sex were transferred into each vial, with up to 7 vials prepared per line. All vials were incubated at 29°C, and the number of dead flies was recorded daily for the next 3 days. This protocol was performed in three independent blocks.

2.5.2 Reproductive output Protocol

The protocol to measure fecundity followed a similar procedure to the mortality protocol, with one key modification. Instead of separating 10 flies of each sex into individual vials, 5 males and 5 females were placed together in a single vial per replicate and incubated at 29°C. This protocol change had to do with controlling density, as we had to place females and males together to ensure all of them mated and to ensure sexual conflict, which is a significant trait of *Drosophila* sexual adaptation (Tennant et al., 2014). Mortality was monitored daily for 10 days, with flies transferred to fresh vials every other day. This ensured conditions conducive to egg-laying and accurate death counting while allowing the next generation to develop at 18°C, preventing interference from the UAS/Gal4 system during development. After each transfer, the vials were stored at 18°C for an additional 15 days, and the number of pupae was subsequently counted.

2.6 Data Analysis

Statistical analyses were performed using R (version 4.4.2) within the RStudio environment (versions 2024.04.3+776 and 2024.09.1+394). The following packages were used for mortality data analysis: Survival and Survival Analysis (Therneau et al., 2000; Therneau, 2024), Survminer (Kassambra et al., 2024), Hmisc (Harrel et al., 2024), Tidyverse (Wickham et al., 2019), Lme4 (Bates et al., 2015),

LmerTest (Kuznetsova et al., 2017), Ggrepel (Slowikowski, 2024), Plyr (Wickham, 2011) , and Dplyr (Wickham, 2022). Survival and clustering plots were generated using the ggsurvplot (Kassambra et al., 2024) function and ComplexHeatmap (Gu et al., 2016) package, respectively. For clustering analysis, the packages Car (Fox & Weisberg, 2019), Lsmeans (Lenth, 2016), ggplot2 (Wickham, 2016), and Survival Analysis (Therneau, 2024) were utilised. For fecundity analysis, we employed the packages Coxme (Therneau, 2024 (2)), Rpart (Therneau et al., 2013), Rattle (Williams, 2011), Partykit (Horthorn & Zeileis, 2015; Horthorn et al., 2006; Zeileis et al., 2008) and ggplot2 (Wickham, 2016) for statistical computations and visualisation.

2.6.1 Mortality Analysis

Analysis of mortality was conducted using the Kaplan-Meier method (*survfit*) to estimate survival probabilities over time. Individuals that survived until the end of the experiment or died from causes unrelated to bacterial infection (e.g., getting stuck in food) were censored. Survival data were further analysed using a Cox proportional hazards regression model (using *coxph* function) to estimate hazard ratios. The hazard ratio represents the relative risk of death in one group compared to another. Specifically, in this study, it quantifies the risk of death in a given OR knockdown line relative to the risk in other OR knockdown lines.

2.6.2 Clustering Analysis

To select high- and low-mortality lines, a clustering analysis was performed. The k-means algorithm was applied to classify OR knockdown lines based on their mortality rates. The optimal number of clusters was determined using the silhouette method to ensure consistency and reliability within clusters. A distance matrix was calculated using the Euclidean method, measuring the square root of the squared differences between corresponding data points. Finally, hierarchical clustering was performed using Ward's method, providing a structured visual representation of relationships among lines.

2.6.3 Reproductive output Analysis

The fecundity of high- and low-mortality lines was analysed by adjusting the number of pupae while also accounting for the mortality observed in each vial, that is, once mortality had occurred, we only considered the number of females in each vial at the time. Pearson correlation coefficient was calculated to assess the relationship between mortality and fecundity, providing insight into potential trade-offs between these traits.

3. Results and Discussion

3.1 Knockdown of different olfactory receptors influences *D. melanogaster* survival upon infection with *Pseudomonas entomophila*

To infer the immune reaction of different knockdown OR lines, including some larval-specific lines serving as a negative control (supplementary materials), we infected them using *P. entomophila* and followed survival for 72 hours. With the results obtained we then drew the survival curves for those lines in both males and females (Figure 3.1).

There was significant variation in survival rates to the bacterium, with noticeable differences between the sexes; females exhibited a lower mortality threshold (Figure 3.1A). Moreover, there is a substantial difference between the highest and lowest survival curves for both males and females. It is also noteworthy that there appears to be a single survival curve among females that stands apart from the rest. Initially, we had anticipated that only a few lines would distinctly differ from the majority, but as illustrated in Figure 3.1, all lines appear to behave somewhat uniquely from one another within that extensive range.

The results obtained show that either knocking down the OR genes influences the behaviour of the immune system on the fly, or that OR knockdown lines vary in fitness. While the factor or factors underlying those results could not be tackled in this thesis, we can still speculate about the possible causes behind this variation.

We propose that one of the underlying causes of this variation could be the up- or down-regulation of AMPs, this hypothesis is supported by Takeuchi et al., 2022 where they found that there was an up-regulation of AMPs in the immune response, where ORs and ORNs were involved. As it has been mentioned before, the role of AMPs is extremely important at all the life-stages of *D. melanogaster*, being mainly produced in the fat body in larval and adult stages and taking a part on several defence mechanisms (Ebrahim et al., 2015; Vanha-aho et al., 2015). Furthermore, AMPs have been shown to be specific towards gram-negative bacteria and to have some specificity against certain types of pathogens (Hanson & Lemaitre, 2020). More recently, other functional roles have been attributed to AMPs, one of such roles is aiding in memory and sleep and in the acceleration of the aging process (Lee et al., 2024; Bandiloo et al., 2018). AMPs are also known to present a swift evolutionary pace, which has recently been predicted to be due to microbiota shaping the host immune response (Hanson et al., 2023). All things considered, AMPs should not be discarded as a possible driving cause behind variation in OR knockdown lines, whether due to their already proven importance in olfaction role as an immunity component, or due to their fast evolution pace.

Another hypothesis that may be behind the differential immune response presented here is the RNAi mechanism itself. It has been shown that there is a discrepancy in the expression of different OR genes in response to temperature changes, with some being up-regulated and others down-regulated (Gomez-Dias et al., 2018; Martin et al., 2011). The findings in question imply that, at higher temperatures, such as the ones used during the infection stage of this experiment, the efficacy of the RNAi mechanism will vary, depending on the expression of the OR genes selected. Under-regulation of genes will cause the RNAi knockdown system to be more efficient, since there will be less transcription produce. The opposite effect will be occurring in up-regulated genes, though, since there will be an increase in the number of RNA transcripts, decreasing the knocking down rate, thus reducing the RNAi mechanism impact. This could have taken place during our experiments, since we infected flies at higher temperatures.

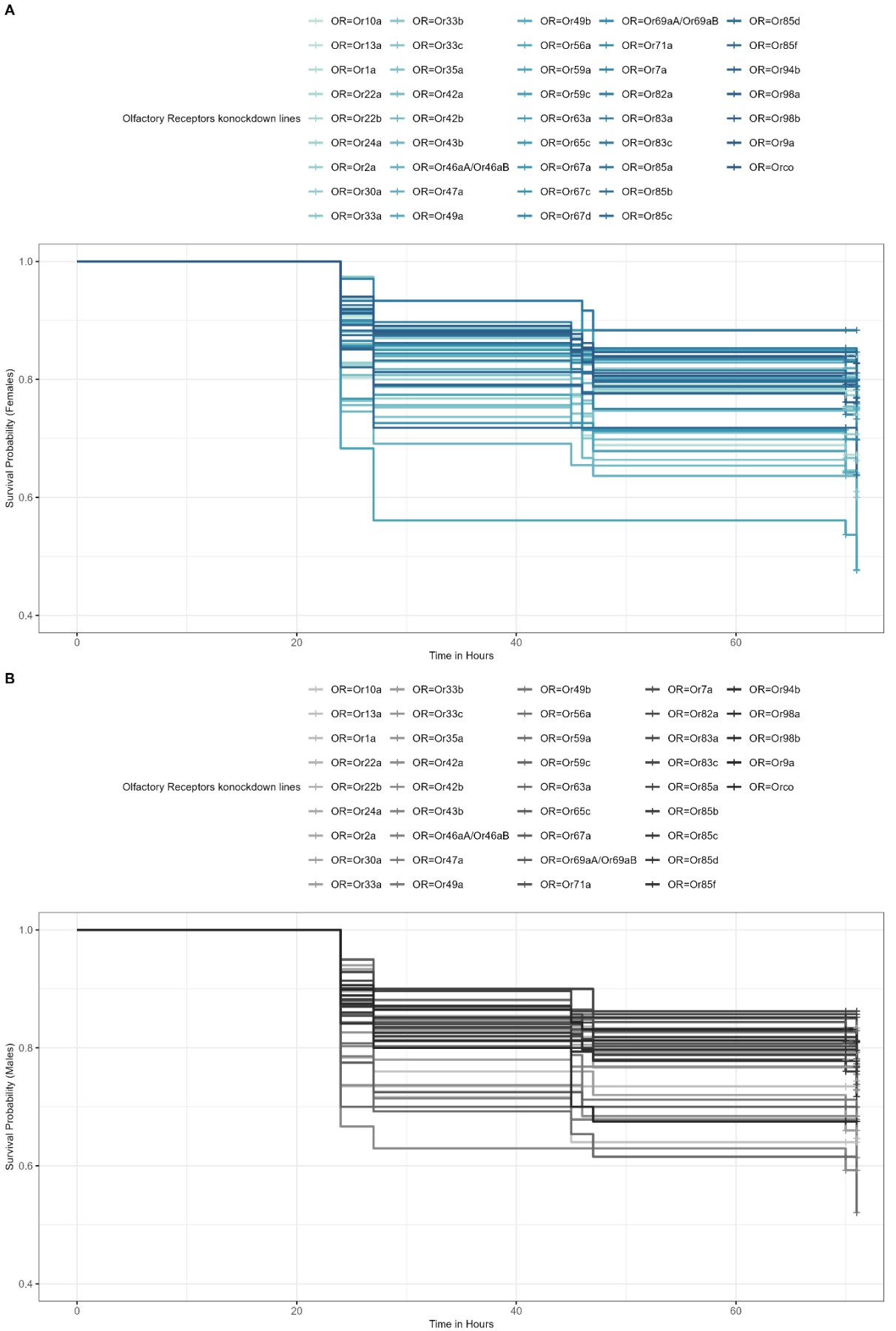


Figure 3.1 - Survival curves depicting survival probability of OR knockdown lines over time. In **(A)** females and **(B)** males. For both plots Y-axis starts at 0.4 survival probability.

3.2 Sexual dimorphism in the response to infection on OR knockdown lines

After fitting a cox proportional hazards regression model to the survival results obtained, we could observe that there were statistically significant differences between males and females for some of the knockdown lines. Our results pointed towards, in some of the knockdowns, males having a higher chance of survival than females (Figure 3.2). Specifically, in knockdown lines Or33b (P=0.012, β=-1.56) and Or33c (P=0.013, β=-1.58) males were associated with a hazard ratio of approximately 0.21 which, in turn, means that males had 79% less chance of dying than females. Similarly, lines Or56a (P=0.004; β=-1.84) and Or7a (P=0.039; β=-1.38), had hazard ratios of 0.16 and 0.25, respectively, meaning that there are 84% and 75% less probabilities of a male dying compared to a female of the same knockdown genotype. Those were the only lines where sex had a significant effect.

There are different biological interpretations that can be inferred from the results obtained, for one it can be argued that it is because of sex-specific immunity; it has been shown before that *Drosophila* as a species show dimorphism in immune responses to several types of pathogens (Belmonte et al., 2020). When reacting to a parasite infection at an elevated density, females tended to decrease less in body size when compared to males (Tseng, 2004). Toll pathway, an important immune pathway, has also been shown to be highly sexually dimorphic when flies are infected (Duneau et al., 2017). Altogether, research seems to reinforce the idea of sex-specific immunity, highlighting the importance of further dwell into such possibility.

Another possible interpretation would be a sex-specific impact in the RNAi knockdown of those genetic sequences. Studies showing similar dimorphism results as a consequence of RNAi mechanism tend to focus on sex specification genes, though they can still provide some insight to this hypothesis, especially when knowledge on olfactive components, particularly olfactory receptors is still vague. In Gotoh et al., 2016 when studying the genetics behind sex determination in *Cyclommatus metallifer*, they found that the knocking down of *intersex (cmix)* and *doublesex (dsx)* genes lead to sex-specific phenotypic alterations. In a study by Ventura et al. (2011), they found that silencing a key sex specification gene in *Macrobrachium rosenbergii*, led to a functional sex reversal in males. While there are no studies correlating such findings with immune response, this hypothesis should not be discarded and should be further researched. As has been pointed out in other studies and reviews, sexual dimorphism affects biological functionality at all levels from genetic to evolutionary (Lee & Chu, 2023; Deegan & Engel, 2019; Kasimatis et al., 2020).

Lastly, it can be argued that the sexual dimorphism observed can stem from behavioural differences inherent to each sex. The behavioural aspect behind olfactory receptors cannot be ignored as it is a crucial part of their functionality. It has been shown that Or47b detects courting hormones in both males and females but only reacts in the presence of male-specific taste pheromones, promoting mating success (Ziegler et al., 2013; Lone et al., 2015). Said findings would support the idea of the gene having an equal expression but having a different effect in males and females, once activated. It has also been shown before that social interaction can also have a key role in immune response (Leech et al., 2019). Flies that did not have contact with the opposite sex were shown to have higher susceptibility to be infected (Leech et al., 2019). This finding shows that behaviour and social environment can have an impact on surviving infections. Future studies should focus on testing how the different functional roles of certain genes and social interactions can cause behavioural dimorphism in immune response.

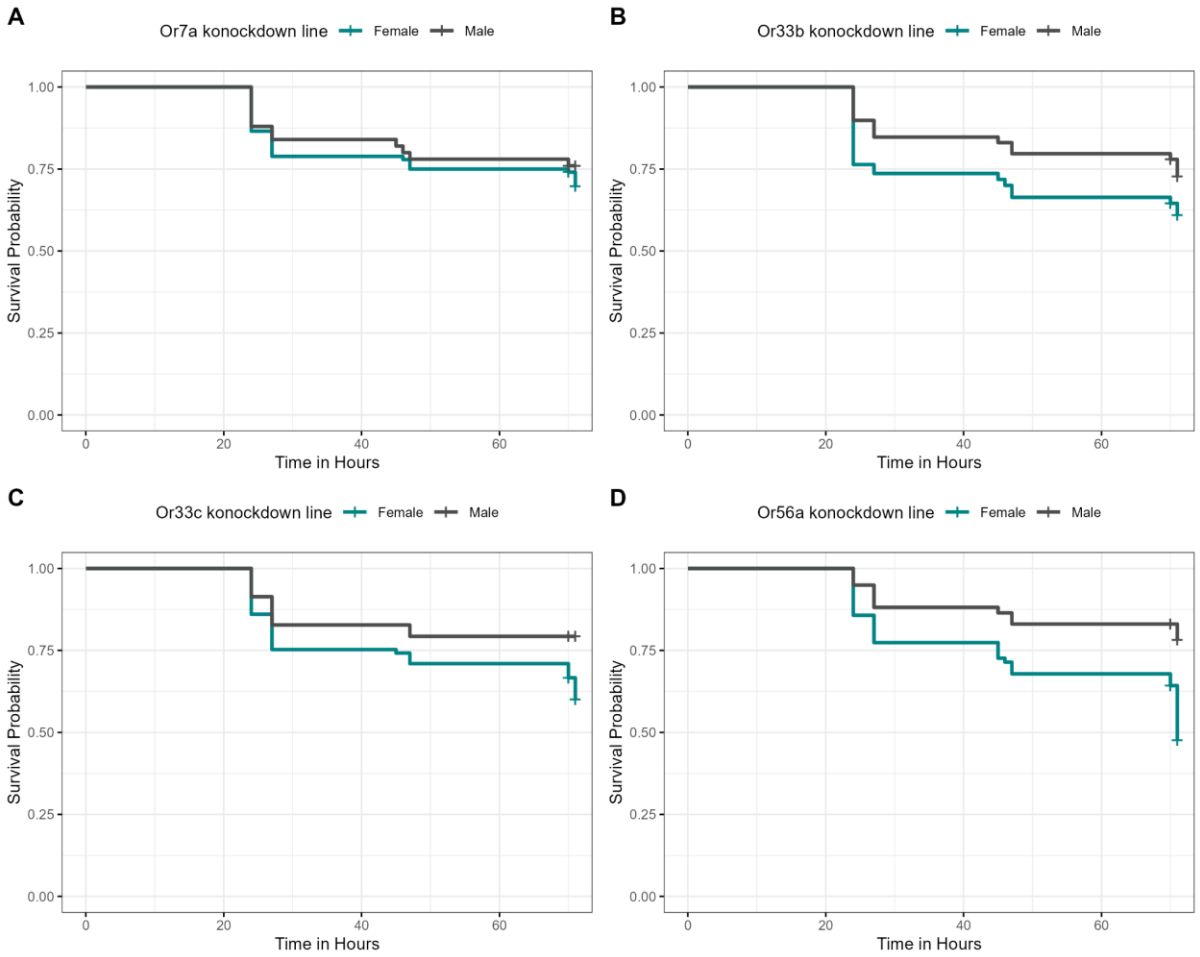


Figure 3.2 - Survival curves of average survival probability in males and females over time in OR knockdown lines that were significantly different. In **(A)** Or7a, **(B)** Or33b, **(C)** Or33c, **(D)** Or56a.

3.3 Clustering analyses reveals knockdown OR lines that had the most extreme responses to the infection

To identify the OR lines with the highest and lowest survival ratios, we conducted a clustering analysis. Mortality data for both males and females were divided into clusters based on average mortality and block-specific mortality. Initially, we aimed to include the top and bottom 20% of mortality lines for both the average and each block. However, this approach was not feasible as no lines met these strict criteria.

We then created four clusters for both sexes and selected lines in the top cluster for average mortality and in the top cluster for at least two of the blocks (Figure 3.3). For low mortality, we chose lines in the bottom cluster for average mortality and in at least two of the blocks. This method worked well for females, but for males, fewer lines met the criteria. To address this, we broadened the criteria for males to include lines in the highest cluster for average mortality and at least within the top two clusters in the blocks.

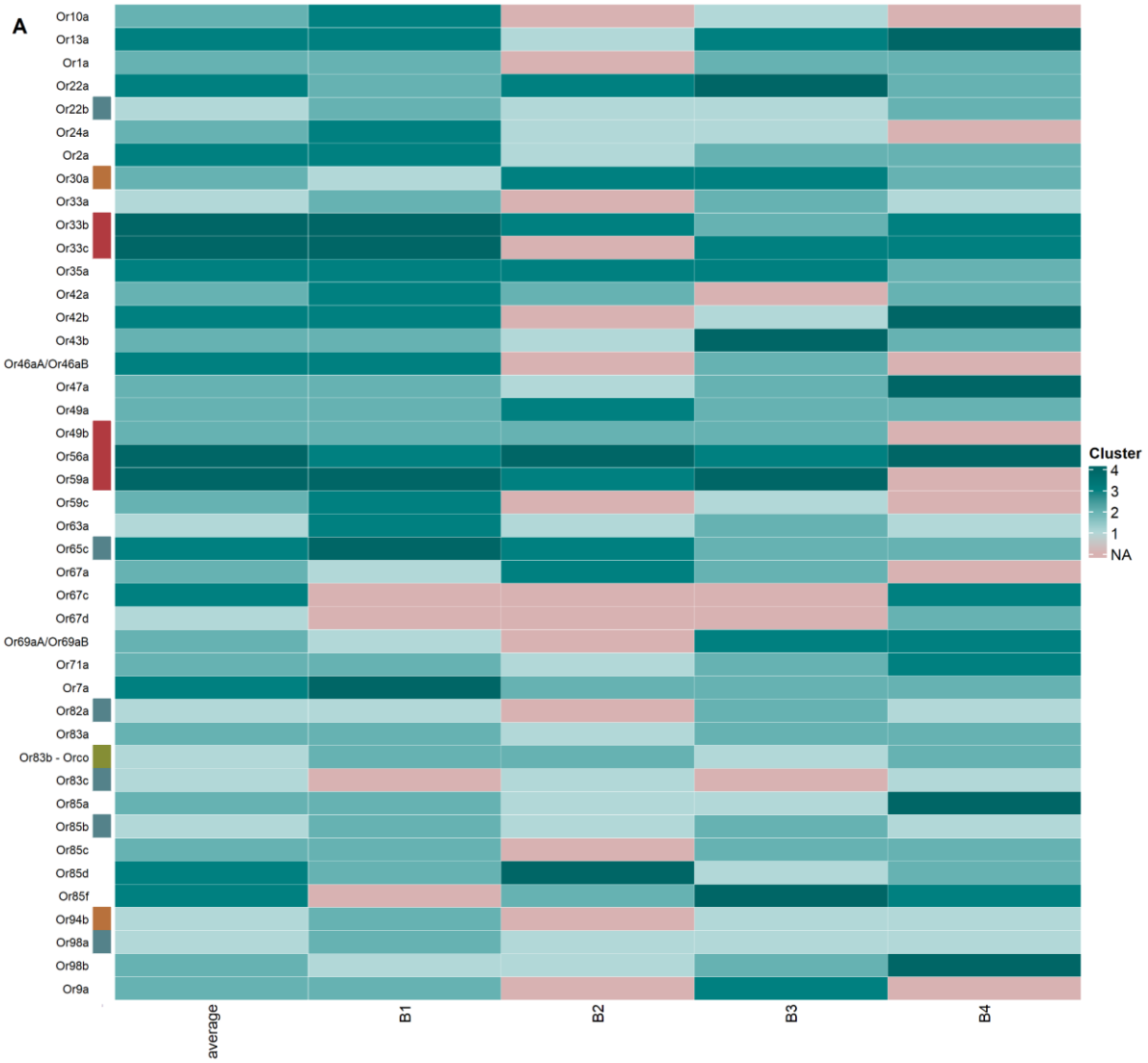
For females (Figure 3.3A), Or59a and Or56a met the criteria for high mortality perfectly, while Or33c, Or33b, and Or65c partially satisfied the criteria. Or33c was in the top cluster for average mortality but only reached the top cluster in Block 1, remaining in the second-highest cluster in the other blocks. Similarly, Or33b was in the top cluster overall but only reached the top cluster in Block 1, while being

in lower clusters in other blocks. Or65c was in the third cluster on average but reached the top cluster in Block 1 and lower clusters in the others. For low mortality, Or82a, Or85b, Or98a, Or22b, and Or83c perfectly satisfied the criteria. Notably, Or59a and Or65c showed overlaps between males and females in high mortality, while Or83c appeared in both sexes for low mortality.

For males (Figure 3.3B), Or59a and Or10a met the criteria for high mortality, with Or65c, Or46aA/46aB, and Or1a narrowly missing the criteria, as they were in the second-highest cluster instead of the top in certain blocks. For low mortality, Or67a, ORCO, Or24a, Or85f, Or83c, and Or85a met the criteria perfectly. The findings reinforce the previously discussed idea of a sexual dimorphism behind the immune response to *P. entomophila*, as the lines selected in the clustering analysis were different in males and females. However, it should still be considered that olfactory receptors, although belonging to the same functional olfactory group, are independent units and, therefore, behave differently from one another. This becomes especially evident when we look at line Or59a, which has been selected as a high mortality line in both males and females, hinting that, for this specific knockdown, there seems to be no sexually dimorphic response.

Interestingly, ORCO was selected as a low mortality line for males but not for females, which emphasizes even more the differences between both sexes. Given that ORCO is extremely important in the activation of ORs, as a mandatory co-receptor, this can have even further implications. Unfortunately, there are no studies on how the expression of ORCO can vary in males and females, thus we cannot really infer what those consequences could be without more studies on it.

From the graphical visualisation and statistical analysis, it is evident that the effect of the block is significant, in future studies a replication of this project should be done in order to reinforce the findings stated here. Moreover, studies focusing on basal mortality of the OR knockdown lines and their effect on the immune response against other types of pathogens should also be included. This would allow us to further confirm our results and also to characterize the impact of infection alone, rather than the impact of both the RNAi knocking down and infection.


A
Or10a
Or13a
Or1a
Or22a
Or22b
Or24a
Or2a
Or30a
Or33a
Or33b
Or33c
Or35a
Or42a
Or42b
Or43b
Or46aA/Or46aB
Or47a
Or49a
Or49b
Or56a
Or59a
Or59c
Or63a
Or65c
Or67a
Or67c
Or67d
Or69aA/Or69aB
Or71a
Or7a
Or82a
Or83a
Or83b - Orco
Or83c
Or85a
Or85b
Or85c
Or85d
Or85f
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1
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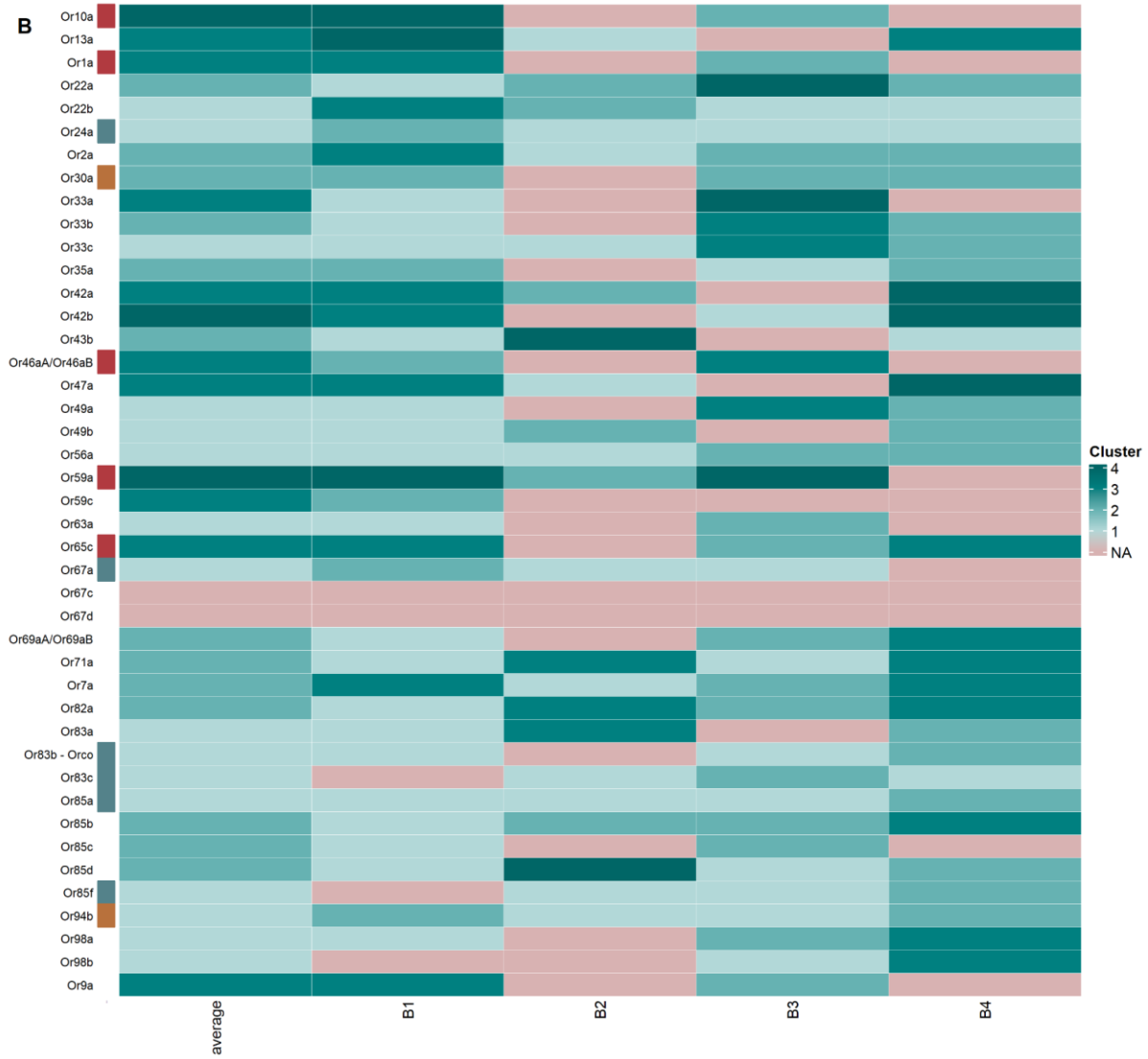


Figure 3.3 - Heatmaps of clustering analysis performed using survival data. Y-axis shows the name of the knockdown OR line, lines that have been selected as having extreme response are coloured next to their name in **(A)** Females and **(B)** Males. Red depicts lines selected as high mortality lines, blue represents low mortality lines, orange shows the larval controls selected for the reproductive output experiment that followed. In females, the ORCO side colour is green. X-axis rows represent the four different blocks performed during the experiment and the average of all the blocks (B1- block 1; B2 - block 2; B3 - block 3; B4 - block 4). The ideal number of clusters was detected to be 4 during clustering analysis (k=4). Clusters are represented in blue, the darker, the higher the mortality. In pink is missing data (NA).

3.4 Reproductive output varies across lines with extreme responses

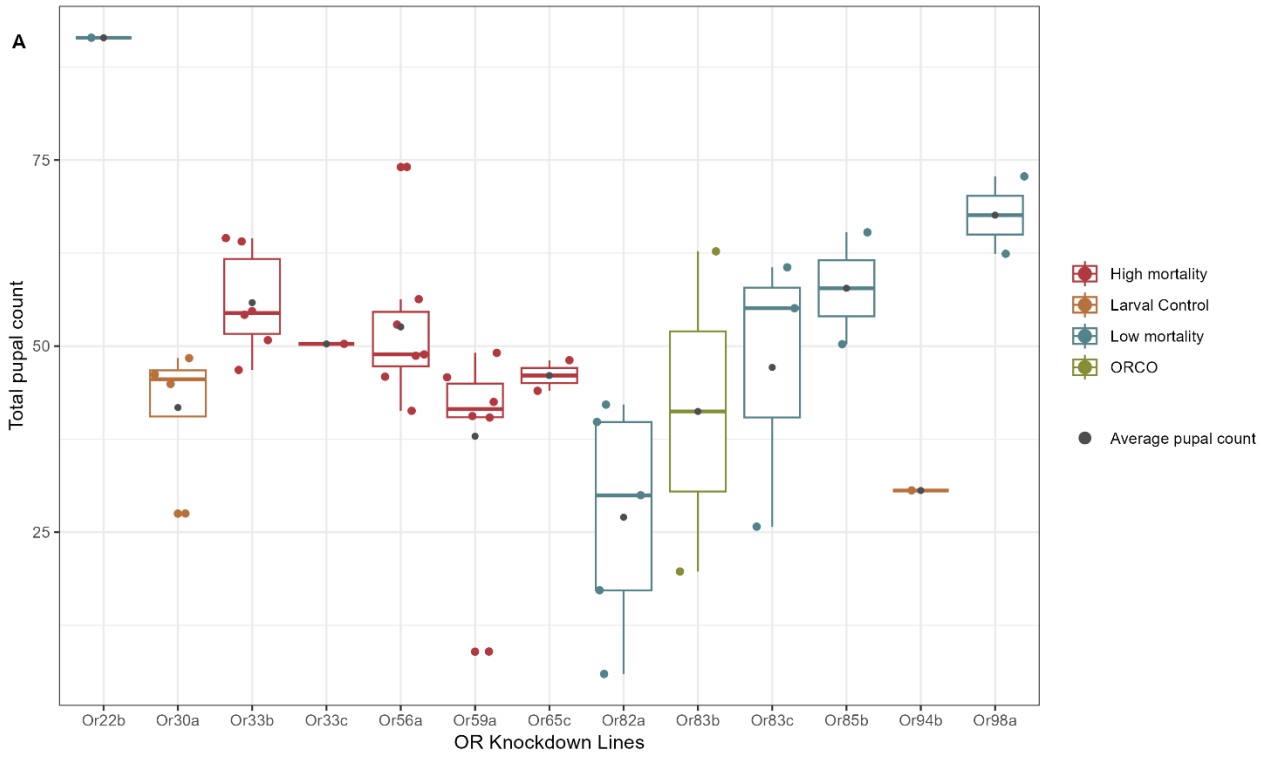
After selecting knockdown lines that had the most polarized outcomes we measured reproductive output from those lines. In this experiment, we measured the reproductive output of extreme response OR knockdown lines infected with *P. entomophila*, we also selected ORCO knockdown line, due to its functional importance as a co-receptor and two larval-specific knockdown lines: Or94b and Or30a (Gomez-Dias, 2018). The reproductive output is the number of pupae, and it was measured by infected females, per vial. Each vial contained 5 females and results have been adjusted to mortality throughout the 10 days of the experiment. Finally, the average reproductive output was calculated, per OR knockdown line.

We found considerable variation in reproductive output across lines as it is shown in Figure 3.4A. Or94b was shown to have the lowest reproductive output with an average of 25 pupae per vial, closely followed

by Or82a with 26 pupae, on average. Or22b line is the one with the highest reproductive output, of 89, however, this line only contained one vial and, therefore, a repetition of the experiment would be necessary to endorse such result. All the other lines are placed in a range of 32 to 54, which shows a not so diversified range of results.

We could also observe that, overall, in most lines the highest pupation occurred on days 7 and 8 post infection (Figure 3.4B). Another interesting thing to note is that high mortality lines Or33c, Or56a, Or59a and Or65c seem to have a lower peak when comparing with their overall reproductive output. This could either mean that the impact of the infection is so high that the recovery time takes longer than in other lines, or that those high mortality lines have a natural lower oviposition rate.

Our results could show a potential survival mechanism, of a diminishing in oviposition rate in favour of allocating resources to immune response to fight off the infection. Once well-being has been restored, and a stabilization of the immune system has occurred, oviposition seems to reach a peak and to later steady again. While there are no studies proving such theory, it is known that environmental hardships, and stress can affect the reproductive fitness and output of both males and females in *Drosophila* (Gefen & Gibbs, 2009; Sadanandappa et al., 2021).



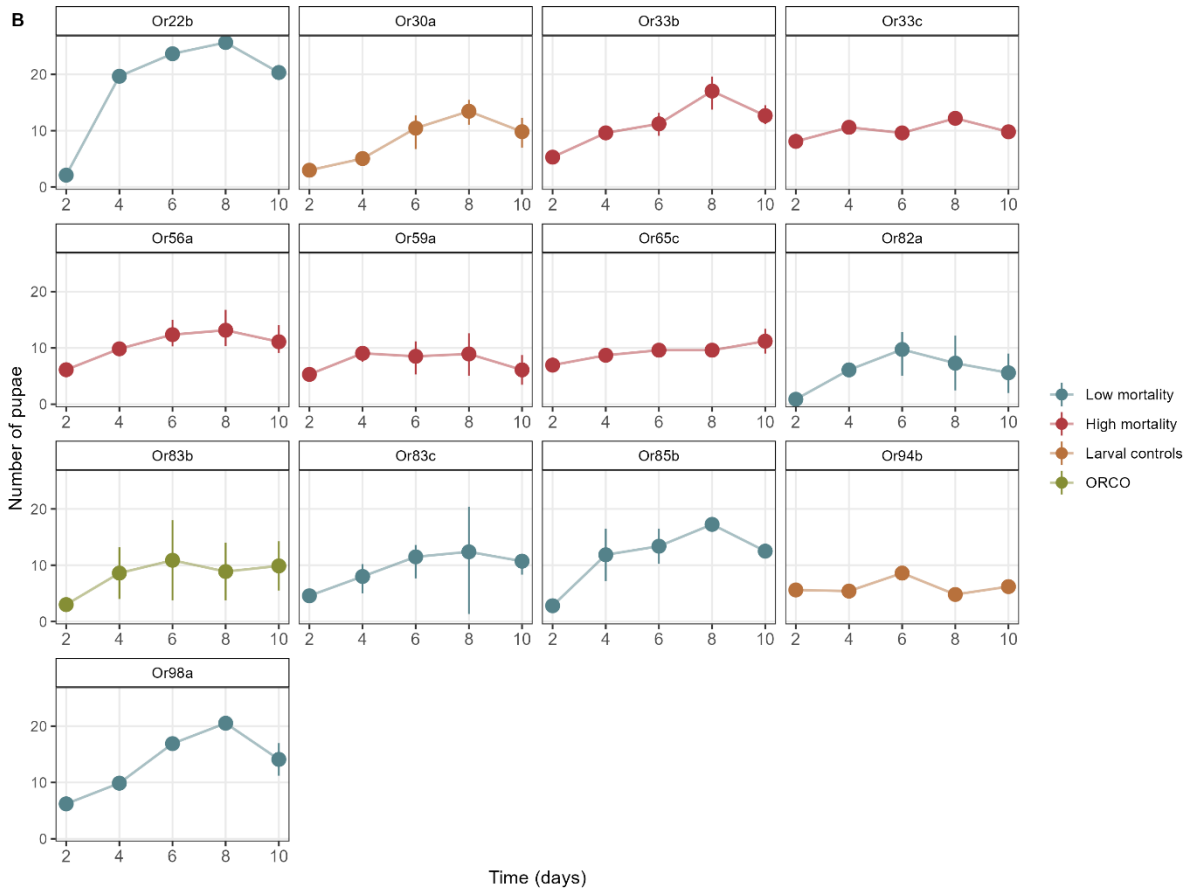


Figure 3.4 – Pupae count in each knockdown line and per day. (A) Boxplot of total pupae count per OR knockdown line after infection. **(B)** Curves for the number of pupae per female in each one of the tested lines over time after infection. Each day contains the data for two days of oviposition, the day of the data collection and the day before. In both plots red represents high mortality lines, blue is low mortality lines, green is ORCO and in yellow are the larval controls.

3.5 No correlation comparing reproductive output and survival

Trying to infer possible trade-offs that may have occurred in the extreme responses, we applied various correlation methods, comparing the survival and reproductive output of the lines tested and traced a linear regression for visual representation (Figure 3.5). Before the experiment we had made some hypotheses that could occur.

First, the energy saving hypothesis whereby we predicted that flies with lower survival would lay more eggs, to save energy for fighting the infection. Kurz et al., 2017, found that the activation of peptidoglycans upon activation of the immune system leads to a slowdown in egg laying. It has been shown before that females that have been to parasitic stress decrease oviposition by retaining mature eggs and eliminating non-mature ones recurring to apoptosis (Sadanandappa et al., 2021). Both studies support the possibility of this hypothesis and reinforce the importance of shedding light on this matter.

Second, the terminal investment hypothesis, a theory that is prominent in literature (Hudson & Vale, 2019; Gulyas & Powell, 2019). In this theory flies with low survival would increase oviposition as a last effort, securing the next generation. This, however, has been shown to depend on the existence of a dietary variation, flies fed a richer diet tend to increase oviposition, while flies in a standard vial tend to invest on egg viability over quantity (Hudson & Vale, 2019). An experiment feeding OR knockdown

lines and controls different diets could confirm whether this hypothesis can be applied in this case or not.

Knockdown behavioural hypothesis, in here we hypothesise that ORs with lower survival cannot perceive the bacteria solution as well as the high survival ones, therefore the high survival lines tend to eat less solution which leads to a lower Bacterial Load, taking to an easier infection to fight off and consequently being able to spend more energy laying eggs. Because there are no studies on this topic, this last hypothesis should be further investigated recurring to metabolic and behavioural studies.

Finally, we idealized that lines with higher survival have a better immune system, therefore controlling and fighting off the infection at a faster pace than the other lines, therefore, spending more energy laying eggs afterwards. From the results we have obtained it seems completely reasonable that this theory could prove to be true in future studies. Not only because we have obtained different behaviours from different OR knockdown lines in all the experiments we present here, including sexual dimorphism, but also the fact that olfactory receptors are independent units, which means that different olfactory receptors can have different effects on the immune system, due to their expression being self-reliant, with the exception of ORCO, which may have an influence in all of the ORs. It has been shown before that genes that do not have a specific role in immunity can regulate innate immune response, such is the case of Osa-Containing Brahma Complex, mainly known as a chromatin restructuring complex (Valanne et al., 2020; Collins and Treisman, 2000).

Our Pearson correlation method, however, indicated no correlation, with a value of 0.19 (P-value=0.55). Therefore, we cannot make any inference on the relationship between reproductive output and survival to infection. Further repetitions of this experiment, with a larger number of samples could prove to be beneficial to confirm the results obtained here or to uncover a hidden relationship between these two components.

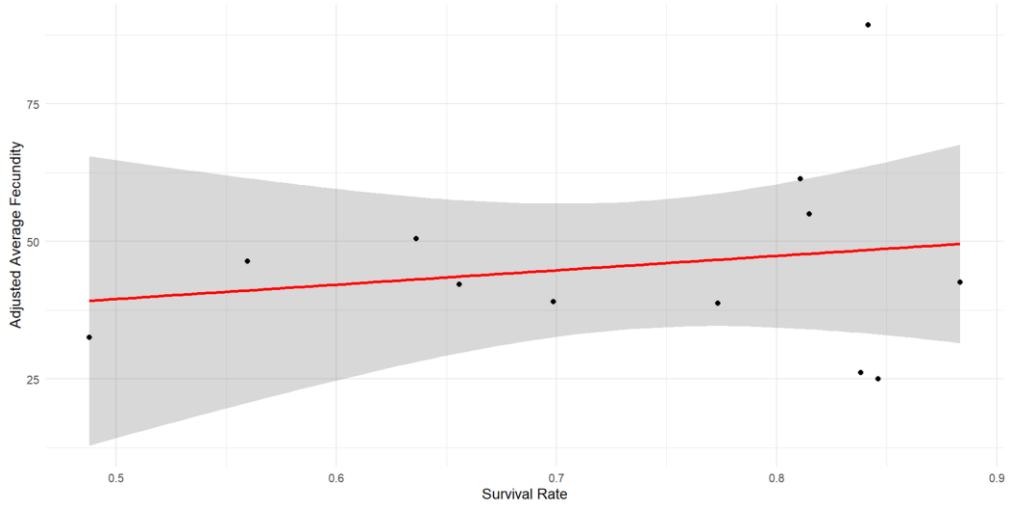


Figure 3.5 - Linear regression of average fecundity (number of pupae) adjusted to mortality and survival rate.

3.6 Contamination in the tubulin line

Once the experiments had been finished, we perceived a contamination had occurred in the tubulin Gal80/Gal4 line used. We could not pinpoint the time of occurrence, thus, the worst-case scenario had to be considered, the loss of the Gal4 component. We reasoned that if this had been the case, the OR

knockdown lines would behave as controls with no knockdown having occurred, that is, they would have all the OR-related genes being transcribed and translated. This would result in the occurrence of false negatives, that is, lines that appear to have had no extreme reaction to the infection but had the Gal4 not been lost, would have. Furthermore, such reasoning also led us to conclude that all of the positive extreme reactions we have had should, therefore, still be considered, as they have a dissimilar response to all the other lines, even if the Gal4 has been lost. In literature, there are no studies discussing in-depth or testing the implications of the loss of the Gal4/Gal80 system in an experimental setup.

3.7 Controlling for RNAi knockdown

A negative control can be defined as an object of study that is given the study treatment but should not produce any result (Földesi, 2024). Here, we used larval specific OR knockdown RNAi lines in a similar way to a negative control. It is known that those ORs are not detected in the adult fly, which means we would expect a non-specific immune reaction from those lines, specifically, we would expect from those knockdown lines a similar reaction to the one of a RNAi control. It can, of course, be argued that while those ORs have not been detected outside of the larval stage, they could still have some other meaningful role in the adult. Naturally, this topic should still be considered as a future object of study, allowing enrichment on the knowledge of the multiple roles of olfaction in *Drosophila*.

The controlling for RNAi knockdown has even further implications in the reproductive output of the protocol used, which had two main variables in study: the effect of knocking-down ORs using RNAi, and the effect of the activation of the immune system through infection. Following the same logic as stated above, here, larval-specific knockdown lines allowed to distinguish the effects originating from the knockdown of ORs.

3.8 Closing remarks

In this MSc thesis we describe the effects of olfaction, namely the knocking down of olfactory receptors, in *Drosophila melanogaster*, in immune response against *Pseudomonas entomophila*. We selected the knockdown lines that showed the most extreme responses. We found no correlation between survival to such infection and reproductive output. Finally, we hypothesise and suggest further research about possible mechanisms and explanations behind the obtained results.

Olfaction and immunity close relation has only recently started to be uncovered. For example, a study by Easwaran and Montell published this year, 2024, has revealed that olfaction and its components are a core element to fecundity after a diapause period, in *D. melanogaster*. Ors have been described to have a key role in boosting immune response against parasitoid wasps in larvae (Madhwal et al., 2020). Chihara et al., 2014 and Takeuchi et al., 2022 also describe the importance of such receptors in aging and neural maintenance. All of this shows how much more there is to discover about the multiple roles of olfactory receptors beyond olfaction.

We conclude that olfactory receptor knockdown lines react differently to infection by *Pseudomonas entomophila*. In some of the infected lines sexual dimorphism was shown, with males and females having different reactions. We selected lines in both males and females where reaction was more extreme, for better and for worse. We also measured the reproductive output in those extreme lines when infected and found enormous variation. Finally, we found that there was no correlation between reproductive output and survival to infection.

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Supplementary materials

Table 1 – Table of the genotypes of Olfactory Receptor lines used and their Bloomington *Drosophila* Stock Center (BDSC) Codes. Table with the full genotypes used in crosses with *tub-GAL80ts; tub-GAL4/TM6* in order to obtain the knockdown olfactory receptor lines used in this study. All genotypes have been described at Perkins et al. (2015).

Olfactory Receptor Line	Genotype	BDSC Code
Or69aA/Or69aB	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05836}attP40	64962
Or71a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05762}attP40	64889
Or82a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06426}attP40	67322
Or83a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06521}attP40	77390
Or83b - Orco	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05293}attP40	63019
Or83c	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23943}attP40	62839
Or85a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06554}attP40	77422
Or85b	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05312}attP40	63038
Or85c	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ24141}attP40	62901
Or85d	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05710}attP40	67812
Or85f	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05307}attP40	63033
Or94b	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ30288}attP40	63720
Or98a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05640}attP40	67802
Or98b	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05397}attP40	66931
Or1a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ21271}attP40	53926
Or2a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ22603}attP40	60385
Or7a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06060}attP40	65185
Or9a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23607}attP40	61981
Or10a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23771}attP40/CyO	62375
Or13a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23682}attP40/CyO	62325
Or22a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05689}attP40	64654
Or22b	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ22718}attP40	60427
Or24a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06423}attP40	67319
Or30a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05455}attP40	66989
Or33a	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23592}attP40	61966
Or33b	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05910}attP40	65036
Or33c	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05934}attP40	77344
Or35a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06048}attP40	65174
Or42a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06035}attP40	65152
Or42b	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23636}attP40	62010
Or43b	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC04963}attP40	57769
Or46aA/Or46aB	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ22886}attP40	60492
Or47a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06472}attP40	67366
Or49a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05600}attP40	64581
Or49b	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06467}attP40	67361
Or56a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05829}attP40	64955
Or59a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06061}attP40	65186
Or59c	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ23243}attP40	61351
Or63a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC06262}attP40	65979
Or65c	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ30023}attP40	62946
Or67a	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS05950}attP40	77356
Or67c	y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMC05789}attP40	64916
Or67d	y[1] v[1]; P{y[+t7.7] v[+t1.8]=TRiP.HMJ30115}attP40/CyO	63549